Quantum Mechanics II — Exercise Sheet 14

The Coulomb Hamiltonian and stability

Variant: Solutions

Exercise 1 (A Gaussian trial energy). Let $H = -\hbar^2\Delta/(2m) - C/r$ in three dimensions, where $C > 0$. For $b > 0$ take the normalized Gaussian $\psi_b(x) = (\pi b^2)^{-3/4} e^{-r^2/(2b^2)}$. (i) Compute $\langle H\rangle_b$ and minimize it over b. (ii) Compare the minimum with the exact lower bound $-\mathrm{Ry} = -mC^2/(2\hbar^2)$.

Solution. The gradient is $\nabla\psi_b = -x\psi_b/b^2$ and $\langle r^2\rangle = 3b^2/2$. Radial integration gives $\langle r^{-1}\rangle = 2/(\sqrt{\pi}b)$. Thus

$$\langle H\rangle_b = \frac{3\hbar^2}{4mb^2} - \frac{2C}{\sqrt{\pi}b}.$$

Differentiating with respect to b gives the minimizing width and energy:

$$b_* = \frac{3\sqrt{\pi}\hbar^2}{4mC}, \qquad \langle H\rangle_{b_*} = -\frac{4mC^2}{3\pi\hbar^2} = -\frac{8}{3\pi}\mathrm{Ry}.$$

Since $8/(3\pi) < 1$, this trial energy is above the exact lower bound, as required.

Exercise 2 (Reduced mass and isotope shifts). A nucleus of mass $M > 0$ and a particle of mass $m_e > 0$ interact through $-C/r$ in their relative coordinate. Take the separation of centre-of-mass motion as given, with reduced mass $m_{\mathrm{red}} = m_e M/(m_e + M)$. Find the ratios of the Bohr length and Rydberg energy to their values at an infinitely heavy nucleus. Expand each ratio through first order in m_e/M.

Solution. The Coulomb scales in the relative Hamiltonian are

$$a_{\mathrm{red}} = \frac{\hbar^2}{m_{\mathrm{red}}C}, \qquad \mathrm{Ry}_{\mathrm{red}} = \frac{m_{\mathrm{red}}C^2}{2\hbar^2}.$$

Dividing by the scales with mass m_e gives

$$\frac{a_{\mathrm{red}}}{a_\infty} = 1 + \frac{m_e}{M}, \qquad \frac{\mathrm{Ry}_{\mathrm{red}}}{\mathrm{Ry}_\infty} = \frac{1}{1 + m_e/M} = 1 - \frac{m_e}{M} + O((m_e/M)^2).$$

The radius ratio is exact, while the displayed energy expansion is a heavy-nucleus approximation.

Exercise 3 (A virial check on an eigenstate). A normalized bound Coulomb eigenstate has energy $E < 0$ and satisfies the lecture's virial identity. (i) Find its kinetic and potential means and $\langle r^{-1}\rangle$ in terms of E and C. (ii) Then use Cauchy–Schwarz to bound $\langle r\rangle$ whenever that mean is finite. Do not assume the hydrogen level formula from Lecture 15.

Solution. For the Coulomb potential the virial relation is $2\langle T\rangle = -\langle V\rangle$. Combining it with $E = \langle T\rangle + \langle V\rangle$ gives

$$\langle T\rangle = -E, \qquad \langle V\rangle = 2E, \qquad \langle r^{-1}\rangle = -\frac{2E}{C}.$$

Cauchy–Schwarz applied to $r^{1/2}\psi$ and $r^{-1/2}\psi$ gives

$$1 \le \langle r\rangle\langle r^{-1}\rangle, \qquad \langle r\rangle \ge \frac{C}{-2E}.$$

The assumption of a finite $\langle r\rangle$ justifies the first of these vectors; the other is controlled by the virial mean.

Exercise 4 (Scaling the domain and the energy). For $a > 0$ let $(D_a f)(x) = a^{3/2} f(ax)$ and $H_C = -\hbar^2\Delta/(2m) - C/r$ on the Coulomb domain $D(H_0)$. (i) Show that D_a is unitary and preserves $D(H_0)$. (ii) Compute $D_a^{-1} H_C D_a$ and express it using $H_{C/a}$. Deduce how the bottom energy scales when C is multiplied by a.

Solution. The substitution $y = ax$ proves preservation of the L^2 norm. The Fourier transform obeys $\widehat{D_a f}(p) = a^{-3/2}\widehat{f}(p/a)$, so finiteness of the fourth momentum moment is preserved in both directions. Differentiation and multiplication give

$$D_a^{-1} H_C D_a = a^2 H_0 - \frac{aC}{r} = a^2 H_{C/a}.$$

Replacing C by aC shows that H_{aC} is unitarily equivalent to $a^2 H_C$. Its bottom energy is therefore a^2 times that of H_C, agreeing with $-mC^2/(2\hbar^2)$.